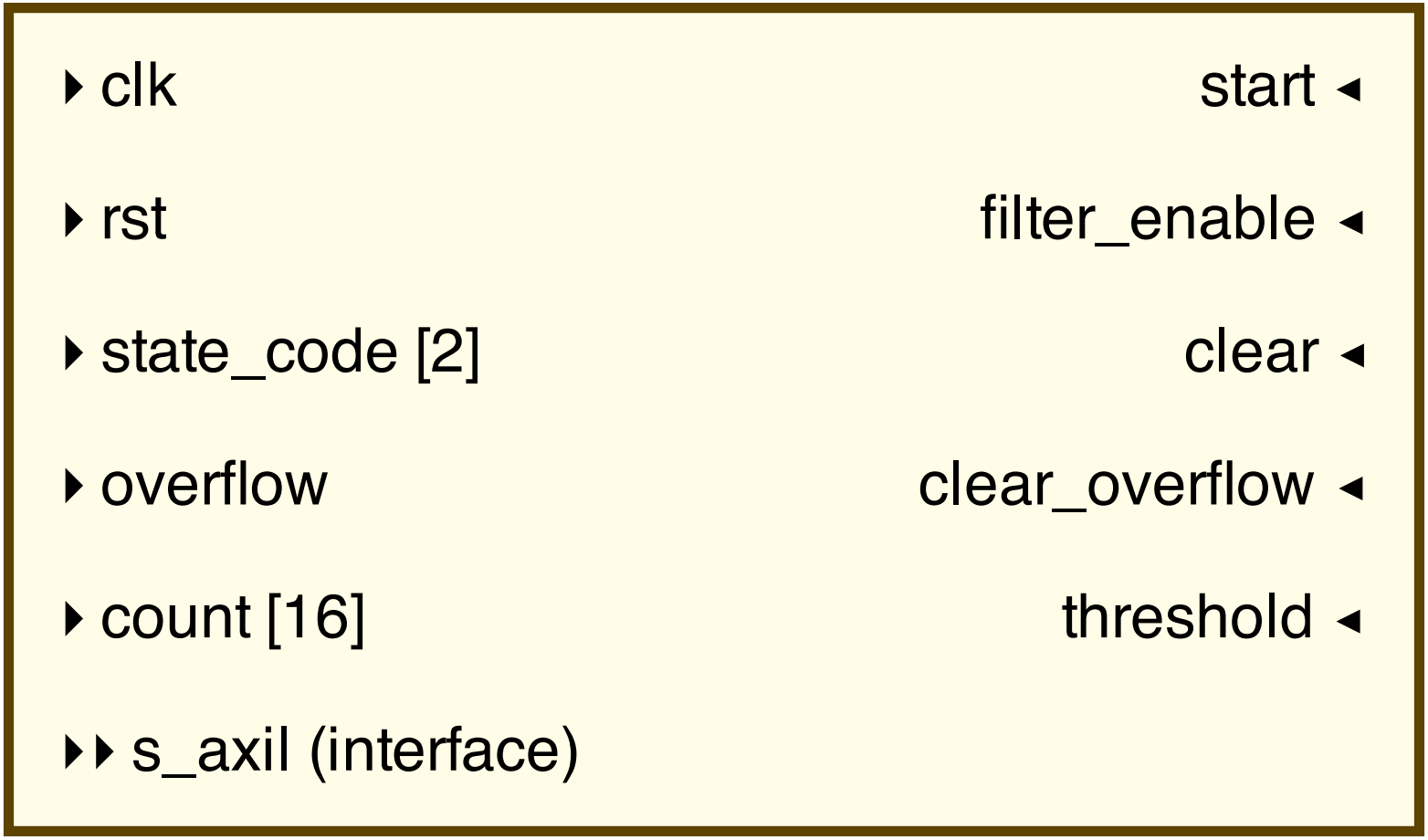
clk
rst
state_code [2]
overflow
count [16]
s_axil (interface)
start
filter_enable
clear
clear_overflow
threshold